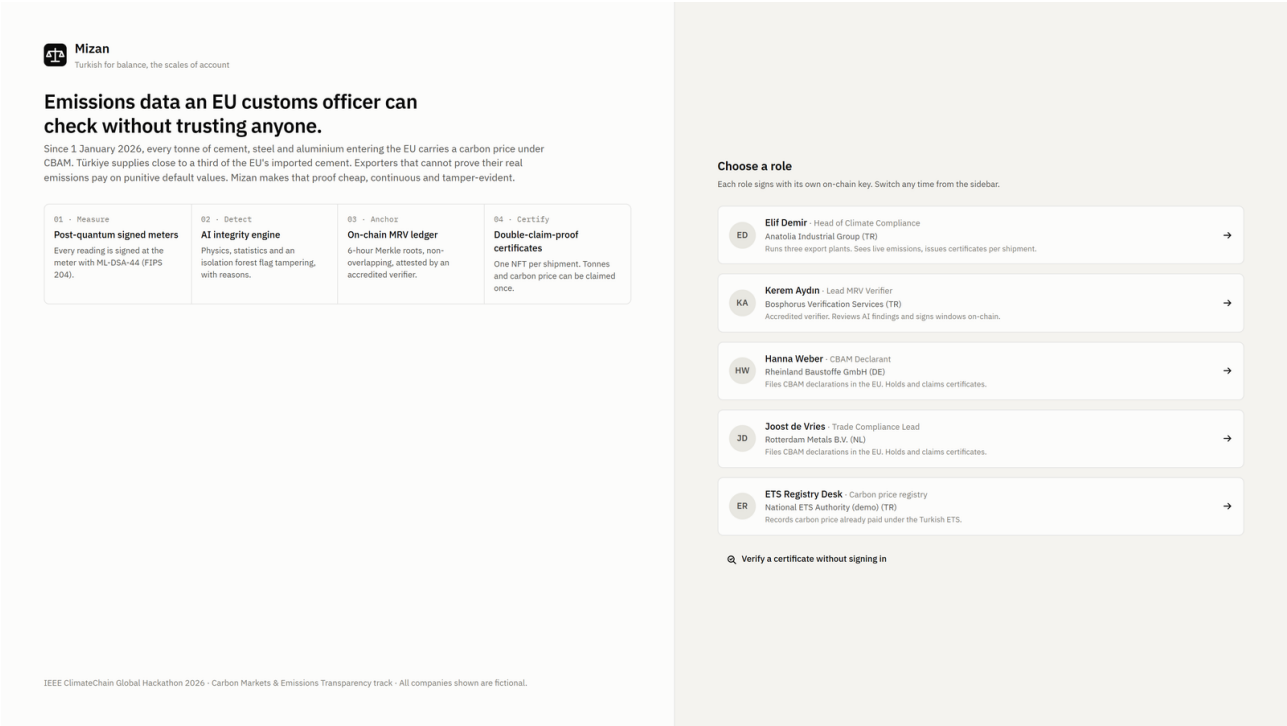
MIZAN

Balance, proven.

Verifiable embedded-emissions accounting for exporters facing the EU Carbon Border Adjustment Mechanism (CBAM)

IEEE ClimateChain 2026 · Carbon Markets & Emissions Transparency track



Live app	https://mizan-159.pages.dev
API	https://mizan-p0rr.onrender.com/api/health
Demo video	https://youtu.be/mW9BZHbX-74
Source code	https://github.com/opeblow/mizan

Pitch: Mizan proves factory emissions for EU CBAM: post-quantum signed meters, AI tamper detection and on-chain anchoring create certificates that can’t be faked or claimed twice.

Inspiration

Since 1 January 2026, every tonne of cement, steel and aluminium entering the EU carries a carbon price under CBAM. Türkiye supplies close to a third of the EU's imported cement, yet most emissions evidence is still a spreadsheet. Exporters that cannot prove their real emissions pay on default values with a mark-up, and carbon price already paid at home can be claimed twice by different importers. We wanted proof that anyone can check and no one can quietly change. *Mizan* is Turkish for "balance", the scales of account.

What it does

- **Measure:** every meter signs each 15-minute reading with ML-DSA-44, the NIST post-quantum signature standard (FIPS 204).
- **Detect:** an AI integrity engine scores every interval with five independent detectors and explains each finding in plain language. A human verifier decides, and that decision retrains the model.
- **Anchor:** readings are sealed every 6 hours into a Merkle tree and anchored on-chain with their emissions and risk score. An accredited verifier attests each window on-chain.
- **Certify:** one ERC-721 certificate per shipment. The contract computes embedded emissions itself and lets each tonne, each euro of carbon price paid in Türkiye, and each CBAM deduction be used exactly once.
- **Verify:** a public verifier checks signatures, Merkle proofs and contract state in the visitor's own browser, without trusting Mizan's server.
- **Plan:** a CBAM exposure planner projects 2026–2034 costs and prices decarbonisation levers.

How we built it

- Three Solidity contracts (MizanRegistry, MRVLedger, EmissionsCertificate) on OpenZeppelin 5, tested with Hardhat.
- TypeScript Fastify API with SQLite; AI integrity engine with an isolation forest written from scratch, robust statistics, physics and mass-balance limits and noisy-OR fusion.
- Meter simulator signing with ML-DSA-44 via @noble/post-quantum.
- React + Vite frontend that re-verifies proofs directly from the chain with ethers.js.
- Backend hosted on Render, frontend on Cloudflare Pages.

Challenges we ran into

- Post-quantum signatures are far larger than ECDSA ones, so batching, storage and request limits had to be designed around them.
- Concurrent on-chain writes collided on nonces; fixed with a per-wallet transaction queue.
- Turning anomaly scores into explanations a verifier can act on.
- Making the browser verify everything itself instead of trusting our API.
- Packaging the chain, API and live meter simulation into one cloud service.

Accomplishments that we're proud of

- A full end-to-end system: meter → AI → chain → certificate → public verification.
- A grid meter altered to report 68% of real load is flagged within one interval, with an estimate of the CO■e it would have hidden.
- Double claims are refused by the smart contract (AlreadyClaimed).
- A value edited directly in the database fails signature, Merkle and on-chain checks.
- 13 contract tests plus unit tests, run in CI on every push.

What we learned

- CBAM’s weak point is trust in the data, not calculating carbon prices.
- Post-quantum cryptography is practical today.
- An AI flag only earns trust when a human can see why it fired and has the final decision.
- A blockchain is most useful as a shared record that prevents double counting.

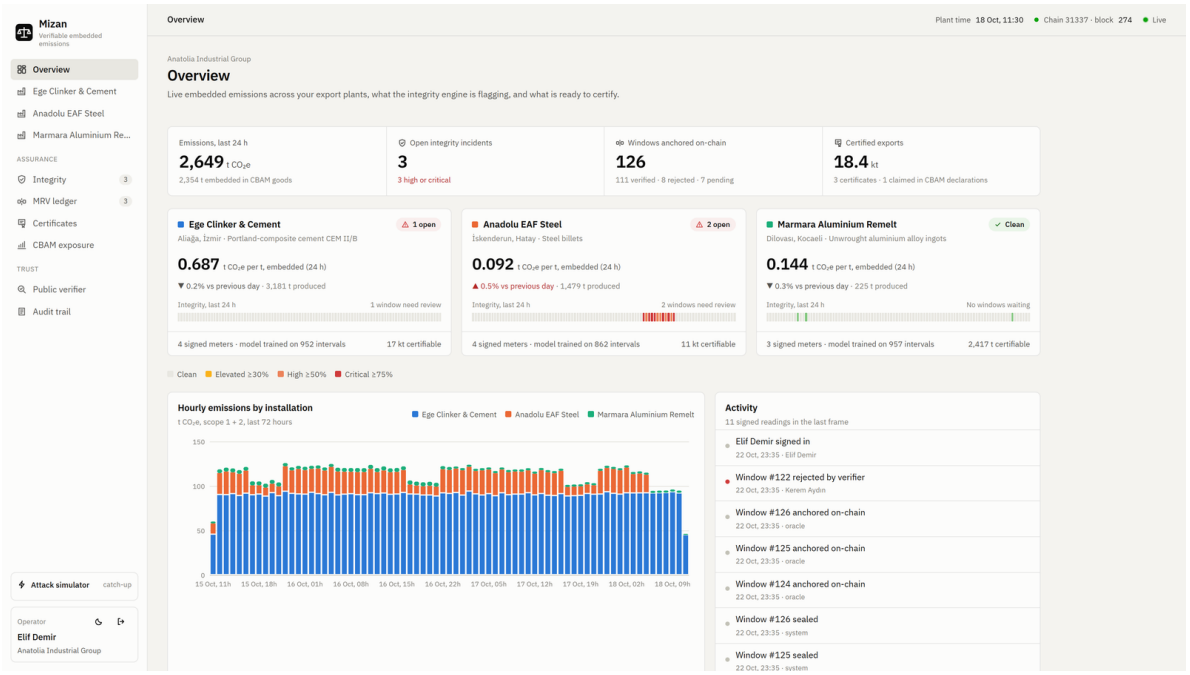
What’s next for MIZAN

- Pilot with a Turkish cement or steel exporter using real meters and hardware-secured keys.
- Public testnet, then a permissioned network for accredited verifiers.
- Map outputs to the EU CBAM Registry reporting format; connect with Türkiye’s ETS.
- Extend to fertilisers, hydrogen and electricity.

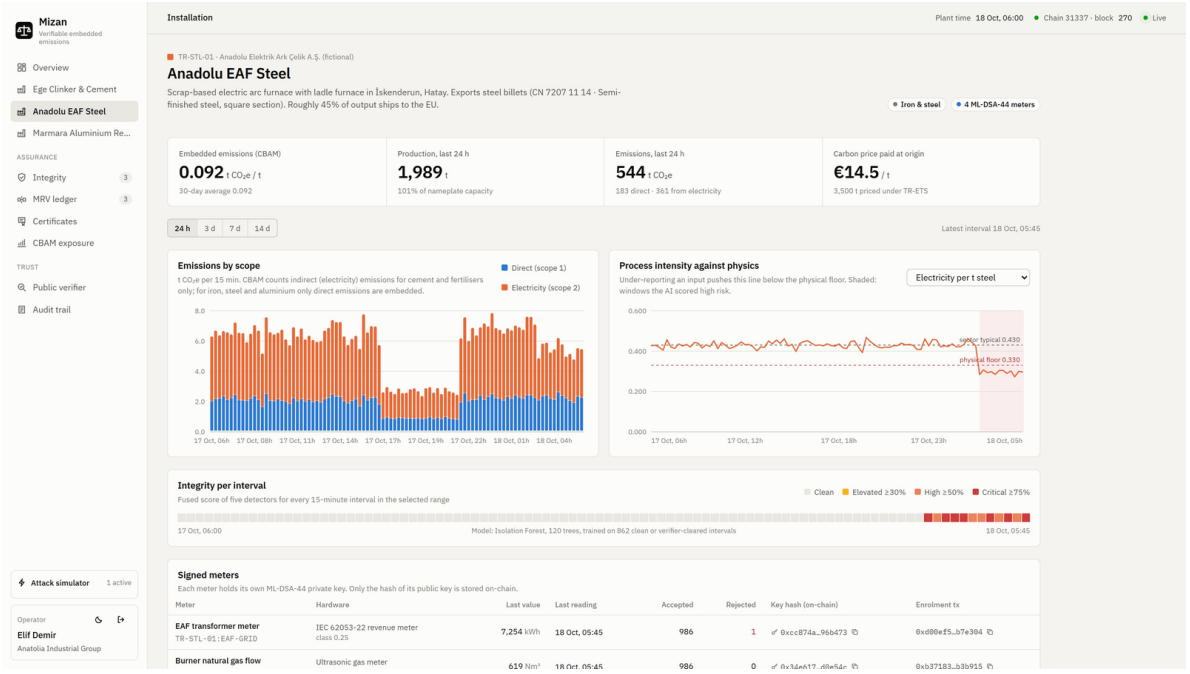
Built with

TypeScript, Solidity, React, Vite, Fastify, Hardhat, OpenZeppelin, ethers.js, SQLite, @noble/post-quantum (ML-DSA-44), Render, Cloudflare Pages, GitHub Actions, Ethereum (EVM), ERC-721.

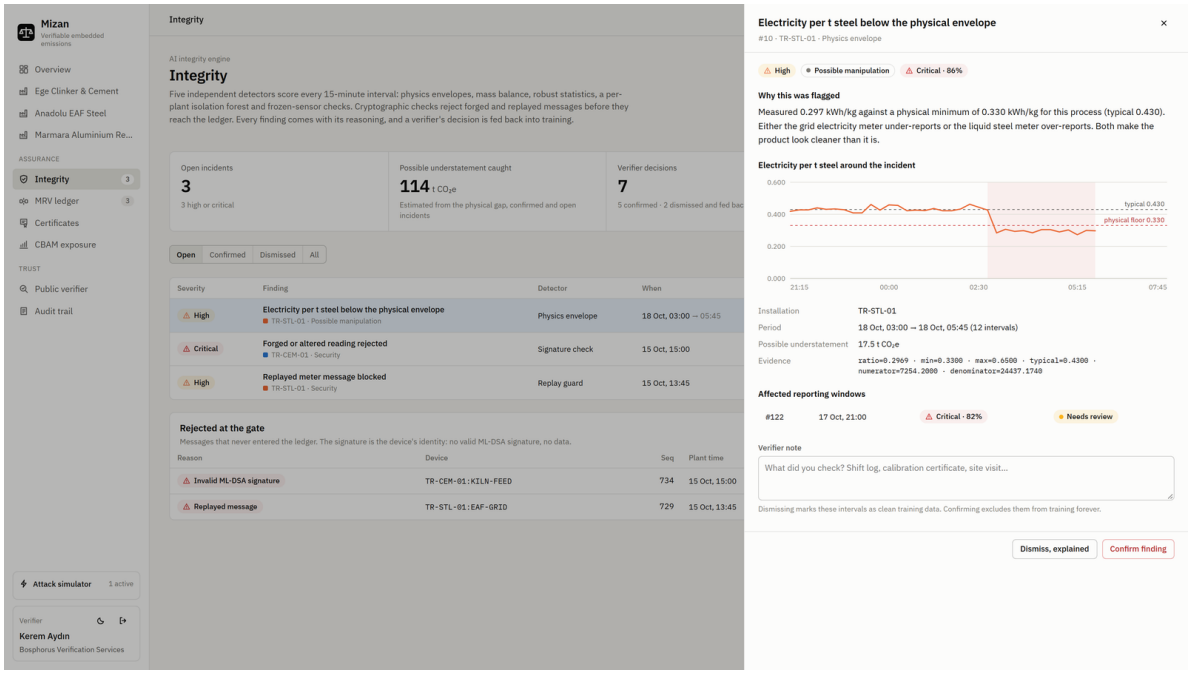
Screenshots (from the running product)



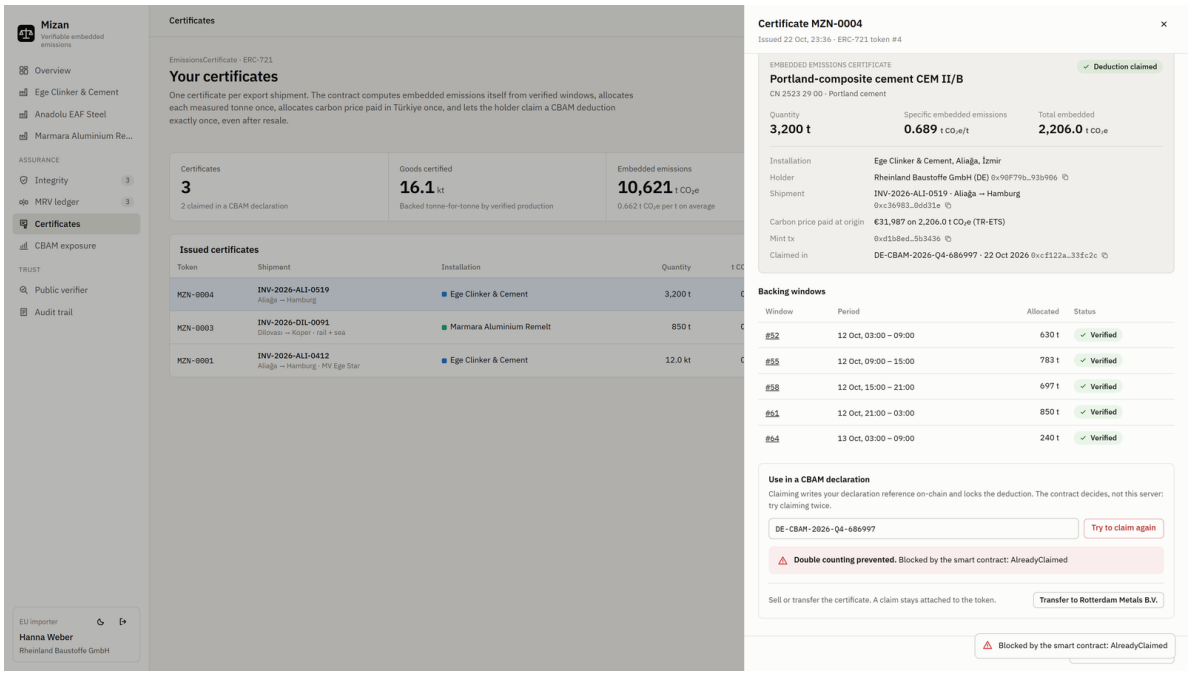
Overview: live embedded emissions per tonne, integrity risk per interval, anchored windows and certified exports.



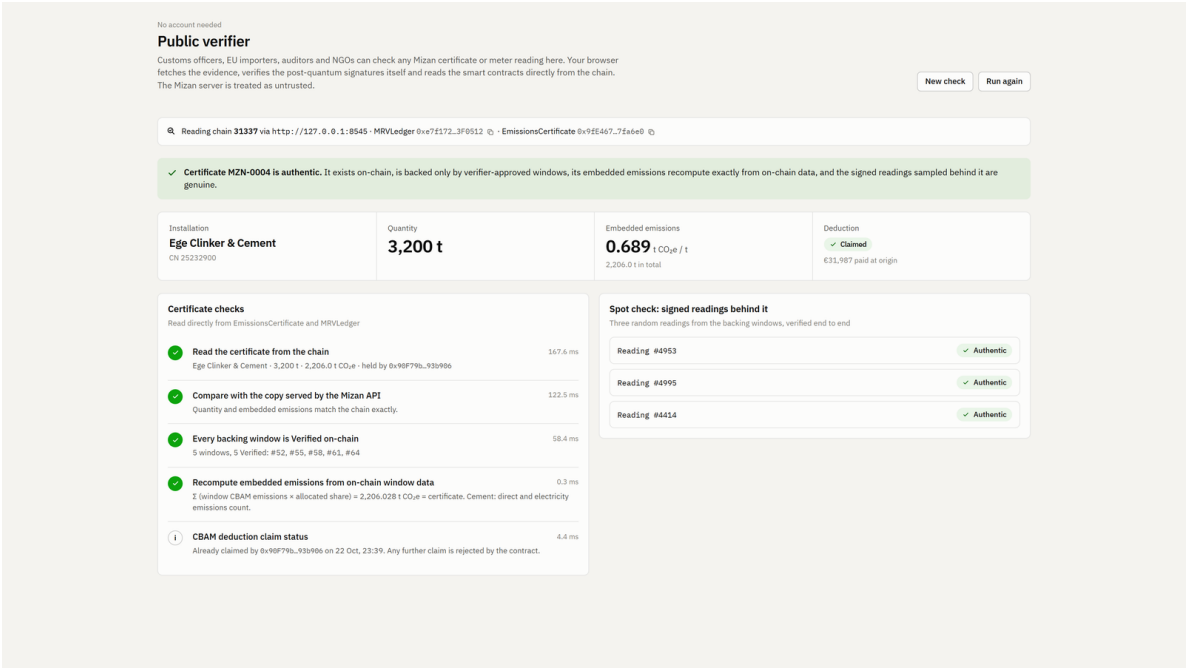
Tampering caught live: a grid meter reports 68% of real load; the risk strip turns red within one interval.



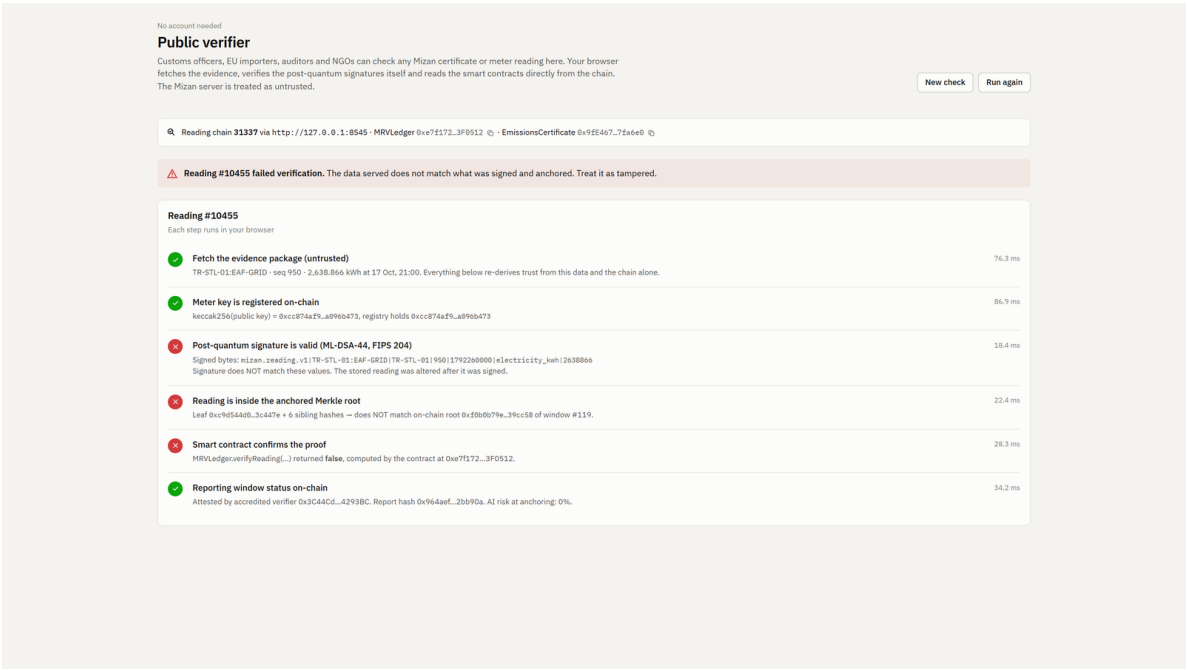
The AI explains: 0.297 kWh/kg measured against a physical minimum of 0.330, likely causes and hidden CO■e.



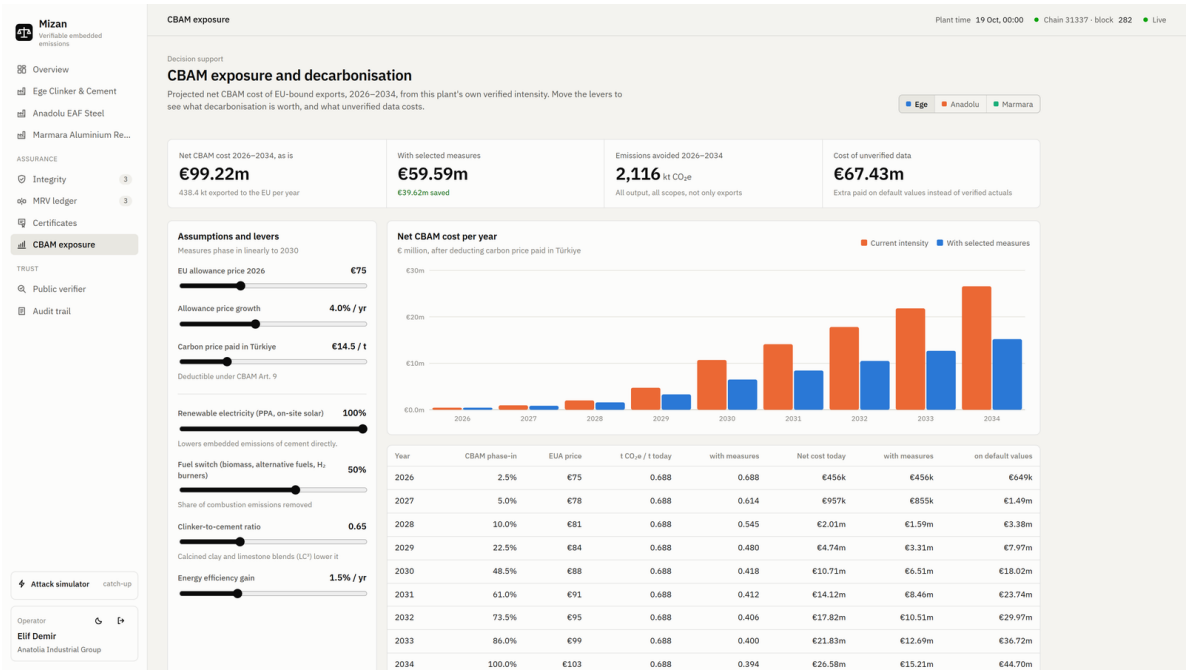
Double counting prevented: a second CBAM claim is refused by the smart contract.



Public verifier: the browser reads the certificate from the chain and re-verifies post-quantum signatures.



Insider edits fail: a value changed in the database fails signature, Merkle and on-chain checks.



CBAM exposure planner: net cost to 2034 with decarbonisation levers.